

Problem 5

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Problem 1 The volume of a doughnut with an inner radius of a and an outer radius of b is

$$V = \pi^2 \frac{(b+a)(b-a)^2}{4}.$$

Find db/da if the volume of a doughnut is $64\pi^2$ and does not change.

Explanation. We have to (implicitly) differentiate

$$64\pi^2 = \pi^2 \frac{(b+a)(b-a)^2}{4}.$$

Before doing this we'll perform a bit of algebra to simplify our calculations:

$$64\pi^2 = \pi^2 \frac{(b+a)(b-a)^2}{4} \implies 256 = (b+a)(b-a)^2.$$

Now we'll differentiate with respect to a :

$$\begin{aligned} 256 = (b+a)(b-a)^2 &\implies 0 = \left(\frac{db}{da} + 1\right)(b-a)^2 + (b+a)2(b-a) \cdot \left(\frac{db}{da} - 1\right) \\ &\implies 0 = \frac{db}{da}(b-a)^2 + (b-a)^2 + 2(b^2 - a^2)\frac{db}{da} - 2(b^2 - a^2). \end{aligned}$$

To finish we solve for $\frac{db}{da}$:

$$\begin{aligned} 0 &= \frac{db}{da}(b-a)^2 + (b-a)^2 + 2(b^2 - a^2)\frac{db}{da} - 2(b^2 - a^2) \\ &\implies -\left(\frac{db}{da}(b-a)^2 + 2(b^2 - a^2)\frac{db}{da}\right) = (b-a)^2 - 2(b^2 - a^2) \\ &\implies \frac{db}{da} = \frac{(b-a)^2 - 2(b^2 - a^2)}{-((b-a)^2 + 2(b^2 - a^2))}. \end{aligned}$$